Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 · UNIT 2 · UE26ME142A Unit 2 Class 16 AVSummary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 2 Class 16 AVSummary

1 UE26ME142A Unit 2 Class 16 AVSummary

Study focus: UE26ME142A_Unit 2_Class 16_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000038 · EV0000039

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Svankitish, Civil Engineering, Department of Science and Humanities, PS University.
- This is class number twenty.
- in class number nineteen, that is previous class, we had discussed how do we go about writing the free body diagram.
- And then we have looked at what are the categories of equilibrium equations that we need to use for a particular type of force system.
- And also we discussed what are the alternate equations of equilibrium that we can use.
- to go out and solve different problems.
- Earlier to this, we had also looked at what are the different types of support that we can get and how do we convert this support or replace the support by their respective support reactions.
- Today we will be solving a few problems from the unsolved problems from the prescribed textbook that is Engineering Mechanics Statics authored by Miriam Craig and Bolton and published by Wiley Publications.
- That is 3 bar 12.
- The statement of the problem goes something like this.
- The 500 kg uniform beam is subjected to the three external loads shown.
- The XY plane is vertical.
- The self weight of the beam is 500 kgs, or I should say it is the mass of the beam is 500 kgs.
- And he says that the x-ray plane is vertical.
- You have a beam which is indicated by O to C and there is a support at O.

- If you recall this is a fixed support, then there are three forces, one is acting at point A, which is acting upwards, it is a point low.
- And the magnitude of that is 1.4 kilonewtons.
- Then we have a movement here which is in the clockwise or no sorry, anti-clockwise direction of magnitude 15 kilonewton meters acting at B.
- And the direction of this force is at thirty degrees to the vertical.
- The first thing that we need to do is to draw a free body diagram.
- Of course the free body diagram is not indicated here.
- But however I'm not be explaining how to draw the free body line.
- Replace the six set support by two reactions.
- Either I can take the sense to be acting upwards or downwards.
- Then you will have the horizontal reaction acting to the left or to the right.
- And then you will have a movement acting at war, which can be clockwise or indicwise.
- If you get a positive answer, the sense that you have adopted is correct.
- If you get a negative sign, that indicates that the sense that you have adopted should be opposite to what you have assumed.
- Then apply the force of 1.4 kilonot at A, apply the moment of 15 kilon meters at B, then you can resolve this force along vertical and horizontal.
- Then the horizontal component of this force will be three sin thirty degrees acting to the left.
- So once you draw the free body diagram, next use the equations of equilibrium.
- In equilibrium chapter, we have to draw the free body diagram.

Example or application from the lecture

- The distance of point A and B is also given, the horizontal distance from O to A is 1.2 meters, from A to B is 1.8 meters, and there is also an inclined point load of magnitude 3 kilonottons, acting at C.
- They have also given you the x axis and y axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 05:00–09:59

Once you draw the free bad item, fifty percent of the problem will be solved.

Key details

- Then use the equations of equilibrium.
- When you are using the equations of equilibrium, remember that if you can use the movement equation first.
- And you choose the point where you want to take the moment as a point where there are unknown that needs to be determined.
- If there are two points where the unknown reactions are acting, then choose the point where the maximum unknown forces are acting as a point where you have to take the moment.
- By doing so, you will be able to get one of the unknown from this equight away.

- Then you can use the other two equation that is sigma f x is equal to 0 and sigma f y is equal to 0.
- So here first I have used sigma m not to be equal to 0 because that is the point where the support reactions are acting.
- Next, 15 kilonewton meters acting in the antipower direction, that would also be positive straight away, add that without multiplying any distance because that is the moment that is there.
- Or you can take the force vertical force that is acting because of this or the component of the 3 cannon force acting around the vertical, which will be 3 cos 30 into the total distance from O to C, which would be 1.2 plus 1.8 plus 1.8.
- Then the self weight of the beam, I forgot to tell you in the when I explained about the free body diagram.
- The self weight also has to be taken.
- The mass of the beam is 500 kgs.
- due to graph D which is G would be the force and that would be acting at the center of the beam which would be 1.2 plus 1.8 plus 1.8 divided by 2, which would be equal to 4.8 divided by 2, which will be 2.4.
- So this is the force that is caused because of the selfate which will be happening the center, that would create a clockwise movement, so it is negative into 4.8 is the total distance.
- The distance one 4.8 by 2.
- Now thousand is to convert the Newton, thus you will get a Newton because Kg into acceleration you to c you will get as 1 newton.
- And equate that to zero from this equation, we get the moment to be 7.56 kilon meters.
- You we have got as positive.
- Next, you take sigma f x equals 0, you have a reaction here.
- I'll assume the reaction at O to be acting to the right, but here the component of the three country force will be acting towards the left.
- And the component is negative, or I can write H0 to be equal to 3 into sine 30 degrees.
- Then we take all the algebraic sum of all the forces acting along the vertical.
- Then we have 1.4 is also acting upwards.
- Then the component of this, since it is acting downwards, on the other side of the equation it becomes 3 into cos 30 degrees.
- Plus we should also take the sulfate of the beam, which is 500 into 9.81.
- Okay, divided by thousand because I have to convert it into kilonewtons.
- With the values by using the equations of equivalent.
- Hope all of you have understood this.
- The next problem that I have chosen is 3 bar 30.
- A former student of mechanics wishes to weigh himself, but as.
- Only to a scale A with capacity limited to four hundred newtons and a small eighteen newton spring dynamometer B with the rig shown.

Example or application from the lecture

- The statement of the problem is given here, which goes something like this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 10:00–14:59

He discovers that when he exerts a pull on the rope, so that B registers 76 Newtons.

Key details

- The scale A reads 268 Newtons.
- What are his correct weight W and mass M?
- We cannot measure anything more than 400.
- And also the maximum capacity of the spring dynamometer is 80 newton, so we cannot measure anything more than 80 newton.
- But when he is doing this experiment, he finds that the scale will read two sixty eight.
- The maximum is four hundred, the scale is reading two sixty eight, it is still possible.
- It is not more than four hundred.
- The rope dynamometer will indicate seventy six newtons, which is also less than eighty, so that is also possible.
- The force FP is seventy six newtons.
- The F of force that is the scale rating is two sixty-eight.
- What we are supposed to find is the weight of the person and the mass of the person.
- The tension that is applying here is 76 newton.
- The weight of the person, which will be W, that is acting at the center of mass of the person.
- W is equal to M into G, then the scale reading which is at the bottom of S shoes, that is 268.
- Then this tension, if you observe, there is tension here which goes here, that this would also be D.
- The same T is continuing here, so there will be T plus T, the same T is continuing here, and thus this is same.
- This would be equal to 40.
- So if I take some of all the forces, F is equal to 0 because there is only vertical forces that is acting.
- 3 bar 15, the statement of the problem is what mass mb will cause the system to be in equilibrium.
- Neglect all friction and state any other assumptions.
- There could be some small fraction that's going to be happening.
- I can resolve this along.
- the plane that is this one and perpendicular to the plane.

Example or application from the lecture

- And since he is holding this, a part of his weight will go here, which is seventy six, that is also given, that is attention.
- And that is what is given written here as 4 into 76 newtons.
- Okay, let's move on to the next problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 15:00–19:59

The component of the sulphate acting parallel to the surface will be 15 to 9.81 into.

Key details

- Then this would also come.
- That is what is the assumption that I am making.
- MA, T da, what is to be find?
- M B solution?
- Assumption all cables attach.
- 2 L B or vertical as I told you, I am assuming that to be vertical.
- That is this tangential axis.
- This is normal axis, T axis is equal to 0, so 2t.
- is equal to 15 to 9.81 sine of this angle 25 I'll get the force to of this the component of the sulphate.
- We will go to the next problem which is three bar nineteen.
- The statements goes like this.
- When the point zero five kg body is in position shown, the linear spring is stretched 10 mm.
- Determine the force P required to break contact at C.
- Complete solutions from A, including the effect of the weight, B neglecting the weight.
- The centroid will be passing through this because it is symmetrical.
- And somewhere along this line, some point it will be acting.
- Next he says the linear spring is structured by 10 mm.
- So if I multiply 10 divided by because it is in M I have to convert it into meters so divided by thousand so we get it as 1.7 into 10 which will be 17.5 into 10 would be the force that is going to be acting which will be pulling B to the left.
- So for this condition, we have to determine what would be this force B.
- Then the self rate, which will be 0.05 into 9.81.
- Into the distance is 0.06, which is in meters, so we get it in Newton meters.
- Then the force here in this spring, which will be 70.5, into this distance is 0.04, which will create a clockwise movement, so that is why it is negative.

Example or application from the lecture

- And that is 0.05 is the mass that is given, so I multiply with 9.81 and.
- That would be the weight of the cell fate of the body that is given, which is 0.05 to 9.81, which will be passing through this line.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 20:00–24:59

Then neglecting the weight, that means we have to remove this.

Key details

- Hope everybody has understood this.
- The next one is 3 bar 20.
- So there is a clockwise movement here which is going to be acting because of the spring, which will be equal to 0.7 newton meters.
- We are again supposed to find out what is the value of P when there is no contact here.
- The self weight also we know, but there's a clockwise movement here, so minus equals 0, so we get P.
- Neglecting the weight, we have to neglect this.
- And take it so we get P as 6.25 newtons.
- We will be continuing in the next class.
- We will solve some more problems in the coming classes.
- My name is Dr.
- Sviving Tesh from PS University.
- We'll see you in the next.

Example or application from the lecture

- This is similar to the previous problem except that instead of a spring that was attached here, linear spring, now they have given you the torsional spring attached at O.
- And they have given that the torsional spring at goes pre-tention so that the so that it exerts a 0.75 N meter clockwise moment.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 2_Class 16_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 16_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 16_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 16_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.

4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.
9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...

C Revision and retrieval practice